



THE UNIVERSITY OF
MELBOURNE

COMP90050 Advanced Database Systems

Semester 1, 2026

Lecturer: Farhana Choudhury (PhD)

Live lecture – Week 11



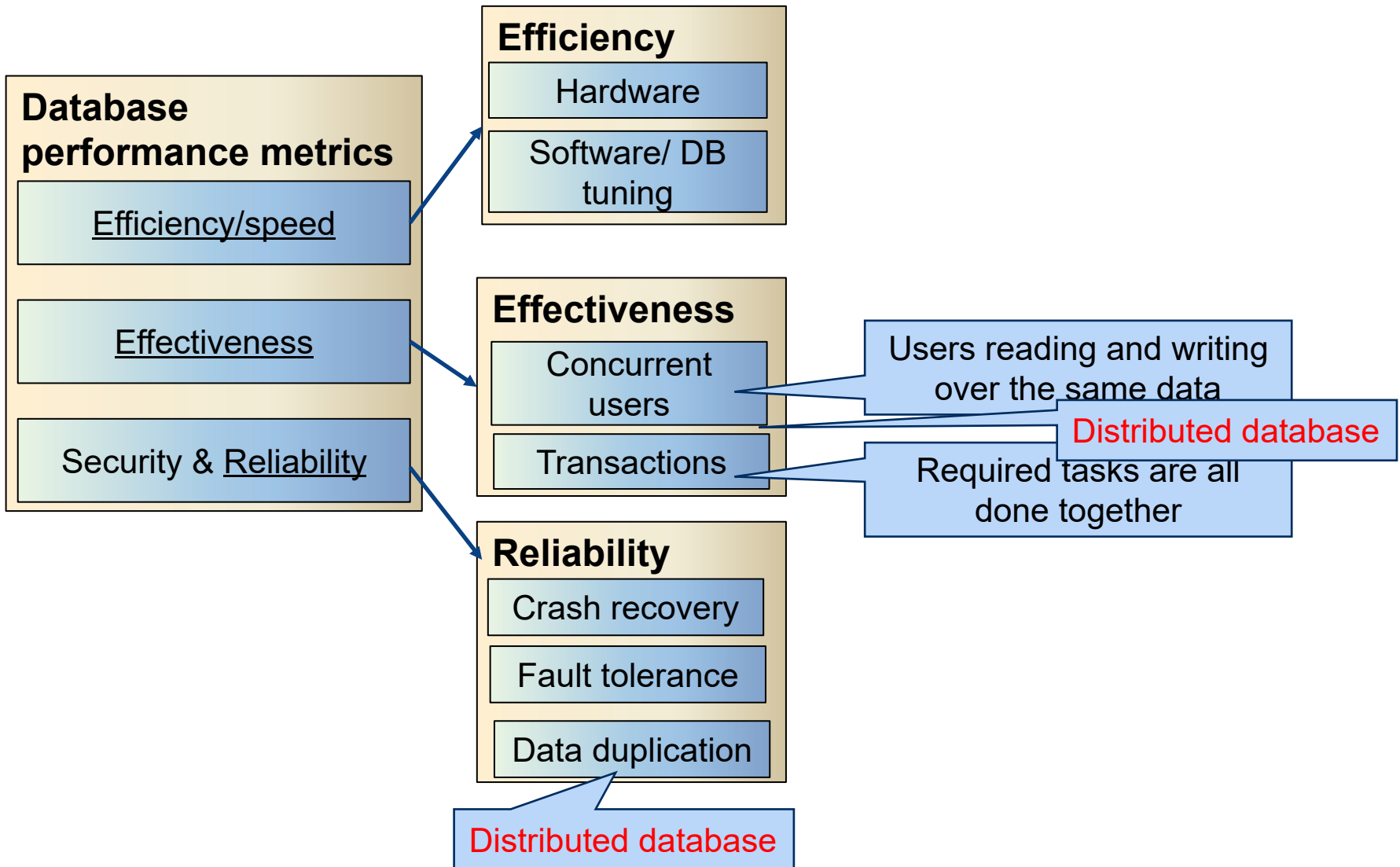


Today's topics

- Activities on distributed databases
- Quiz questions
- Q/A: Project, final exam



Core Concepts of Database management system





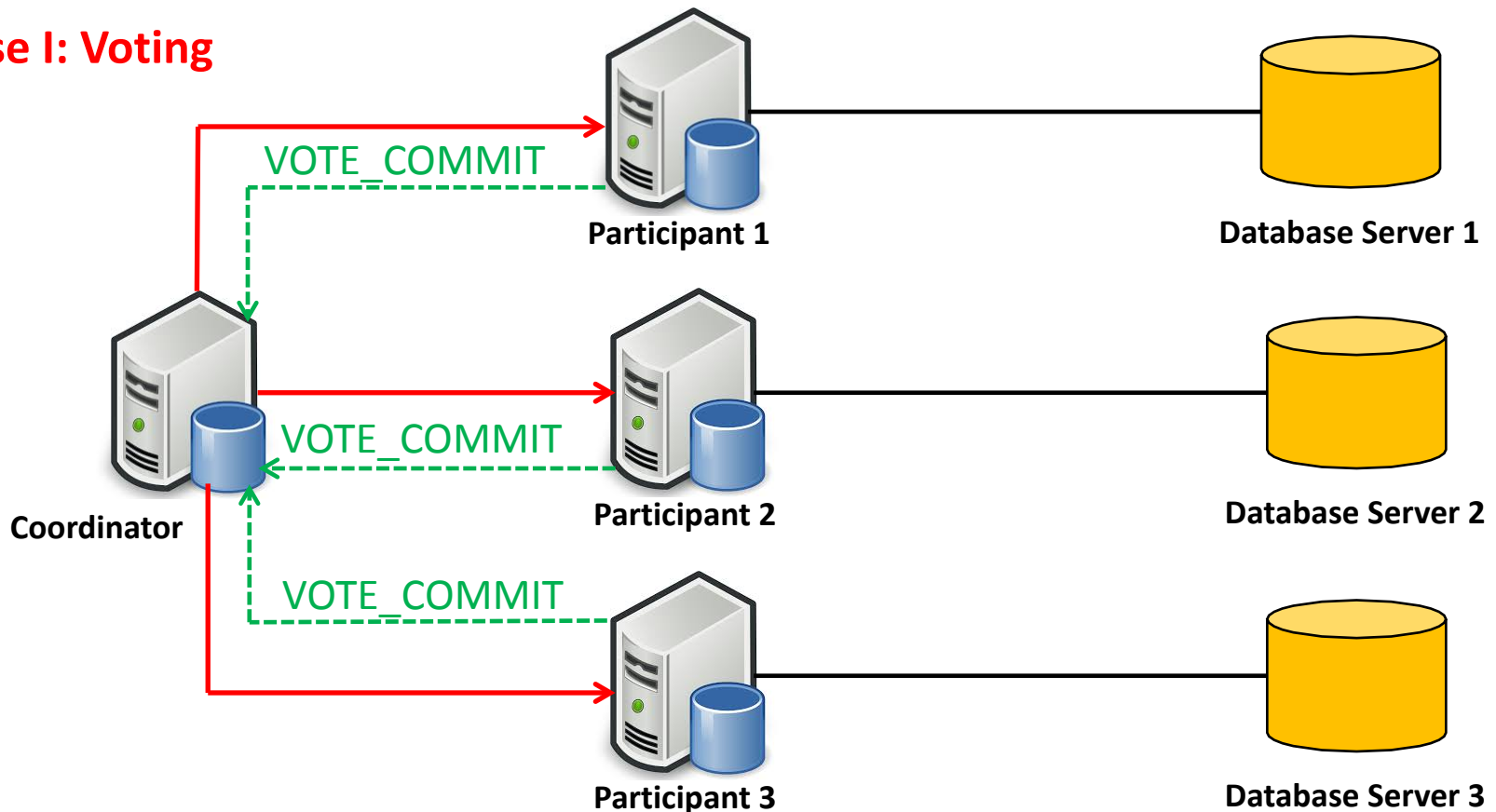
DBMSs in the era of Networking

For distributed transaction processing:

- Atomicity: two-phase commit protocol

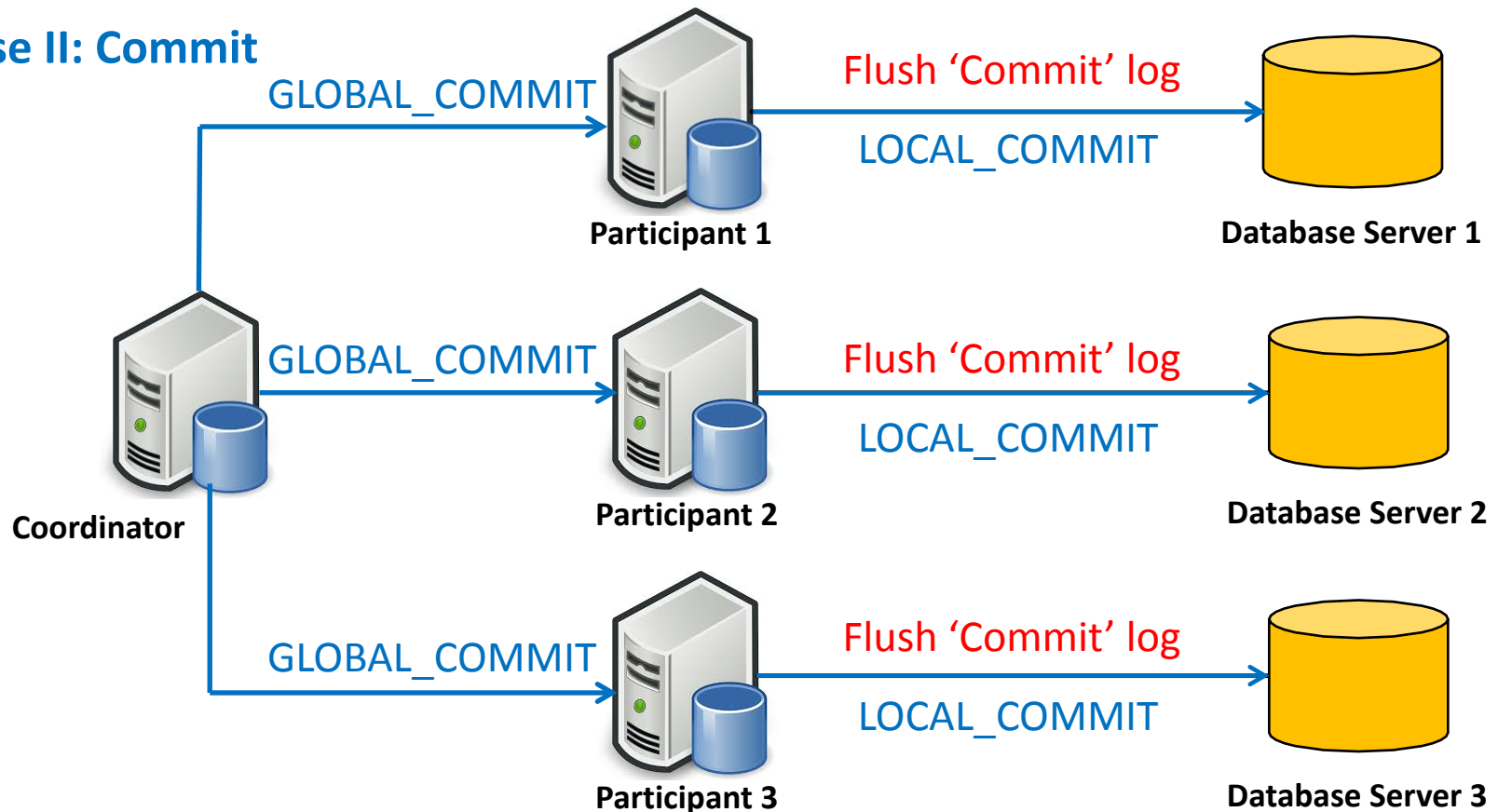
Atomicity in distributed DB: Two phase commit protocol

Phase I: Voting



Atomicity in distributed DB: Two phase commit protocol

Phase II: Commit





Atomicity in distributed DB

In a distributed DB, a transaction T is being executed as T1 in node D1 and as T2 in node D2 (T1 and T2 are parts of T). Both T1 and T2 successfully finished their operations in their own nodes, but D2 lost connection with the coordinator, and couldn't send the 'yes' vote in phase 1. Can T successfully commit?

Time for a poll –

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DBMSs in the era of Networking

For distributed transaction processing:

- Atomicity: two-phase commit protocol
- Concurrency:
 - Lock release
 - Timestamping
 - Optimistic locking



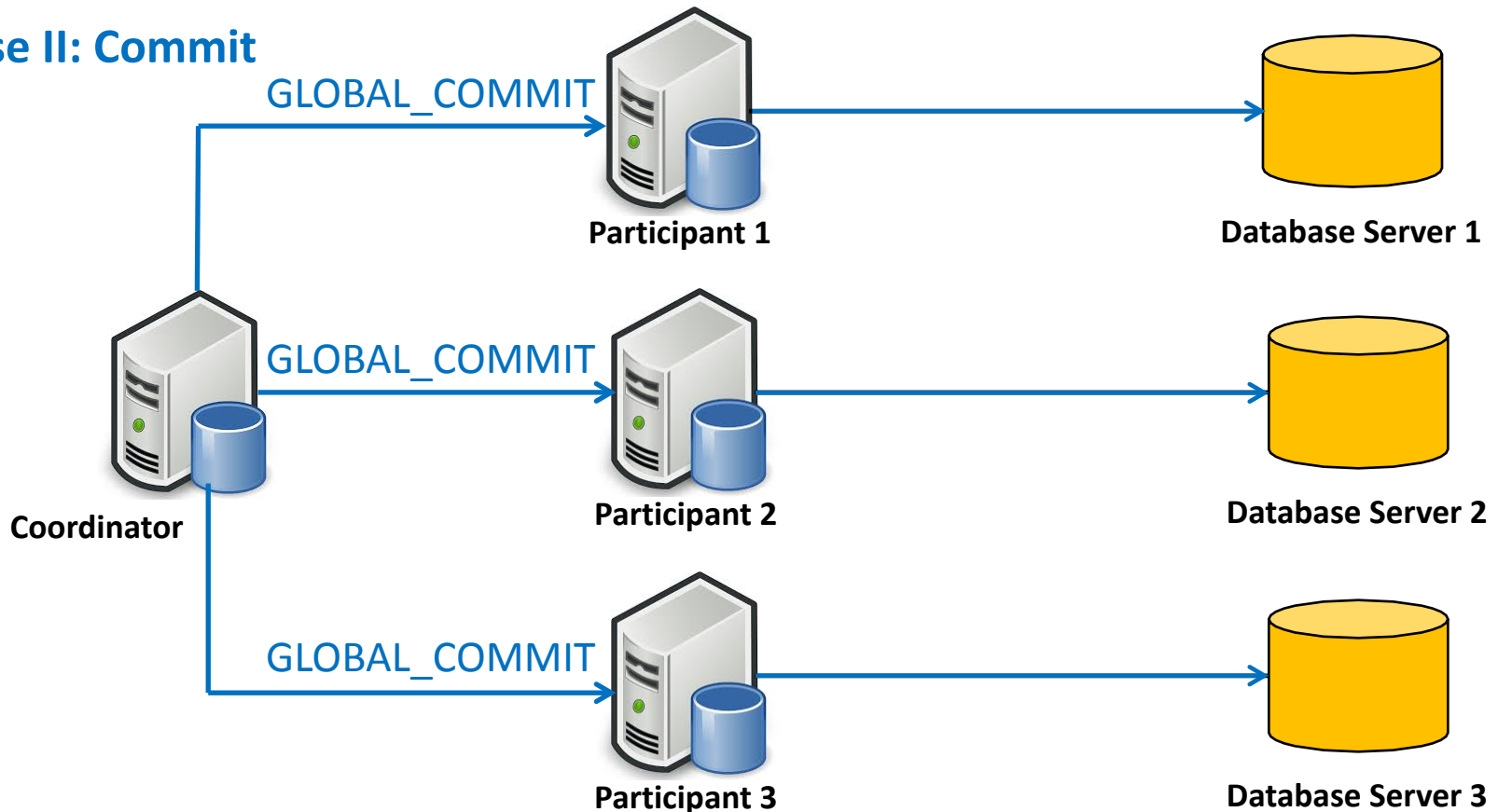
Concurrency in distributed DB

A local lock manager cannot release any locks until it knows that the transaction has been committed or aborted at all the servers involved in the transaction.

When/how will a node know when to release the locks of a transaction?

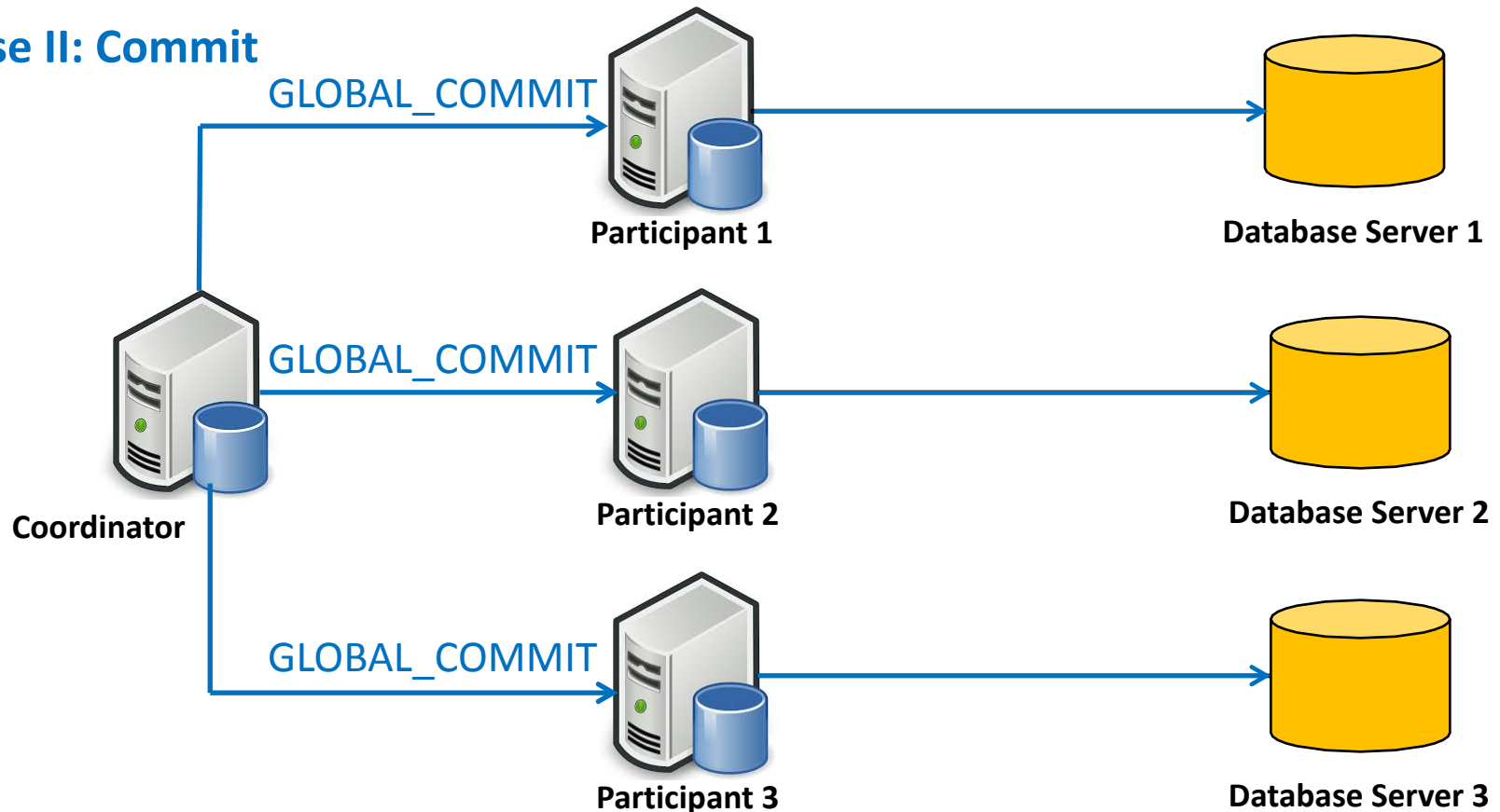
When/how will a node know when to release the locks of a transaction?

Phase II: Commit



When/how will a node know when to release the locks of a transaction?

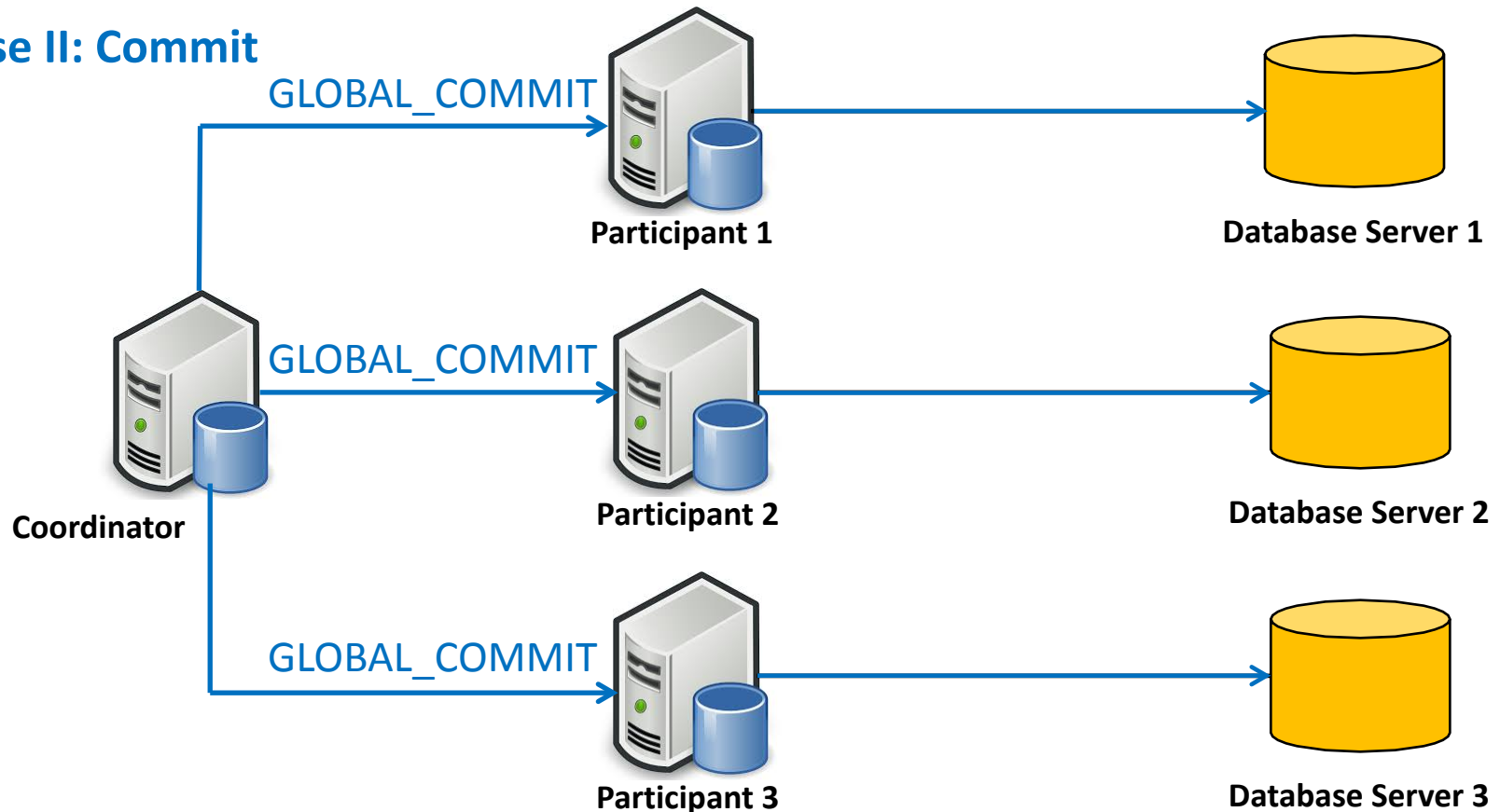
Phase II: Commit



What if a sub-transaction aborts in a node?

When/how will a node know when to release the locks of a transaction?

Phase II: Commit



What if a sub-transaction aborts in a node? – safe to do in phase I



DBMSs in the era of Networking

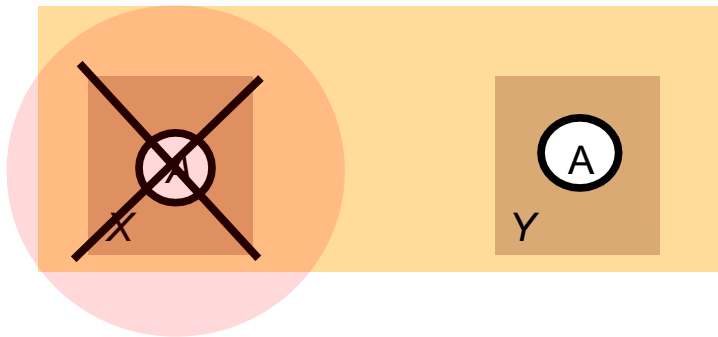
For distributed transaction processing:

- Atomicity: two-phase commit protocol
- Concurrency:
 - Lock release
 - Timestamping
 - Optimistic locking
- Data replication
 - Rule for one-copy serializability
 - CAP

Available copies replication rule

Before a transaction commits, it checks for failed and available servers it has contacted, the set should not change during execution:

- E.g., T would check if X is still available among others.
- We said X fails before T 's *deposit*, in which case, T would have to abort.
- Thus no harm can come from this execution now.





Available copies replication rule

In a distributed DB, a transaction T only needs to read an object o_1 . o_1 is replicated in the nodes D_1 , D_2 , and D_3 of this distributed system. There is another node D_4 , but D_4 does not store any copies of o_1 .

When T started its execution, the nodes D_1 , D_2 are available, and the other nodes are unavailable. When T commits, the nodes D_1 , D_2 , D_4 are available and D_3 is unavailable. Can T successfully commit?

Time for a poll –

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The CAP Theorem

- The limitations of distributed databases can be described in the so called the **CAP theorem**

CAP theorem: any distributed database with shared data, can have at most two of the three desirable properties, Consistency, Availability, or Partition tolerance

Different design choices based on CAP theorem: Examples

- Different data may require different consistency and availability
- Example:
 - Shopping cart: high availability, responsive, can sometimes suffer anomalies/inconsistencies
 - Product information need to be available, slight variation in inventory is sufferable
 - Checkout, billing, shipping records must be consistent

Different design choices based on CAP theorem: Examples

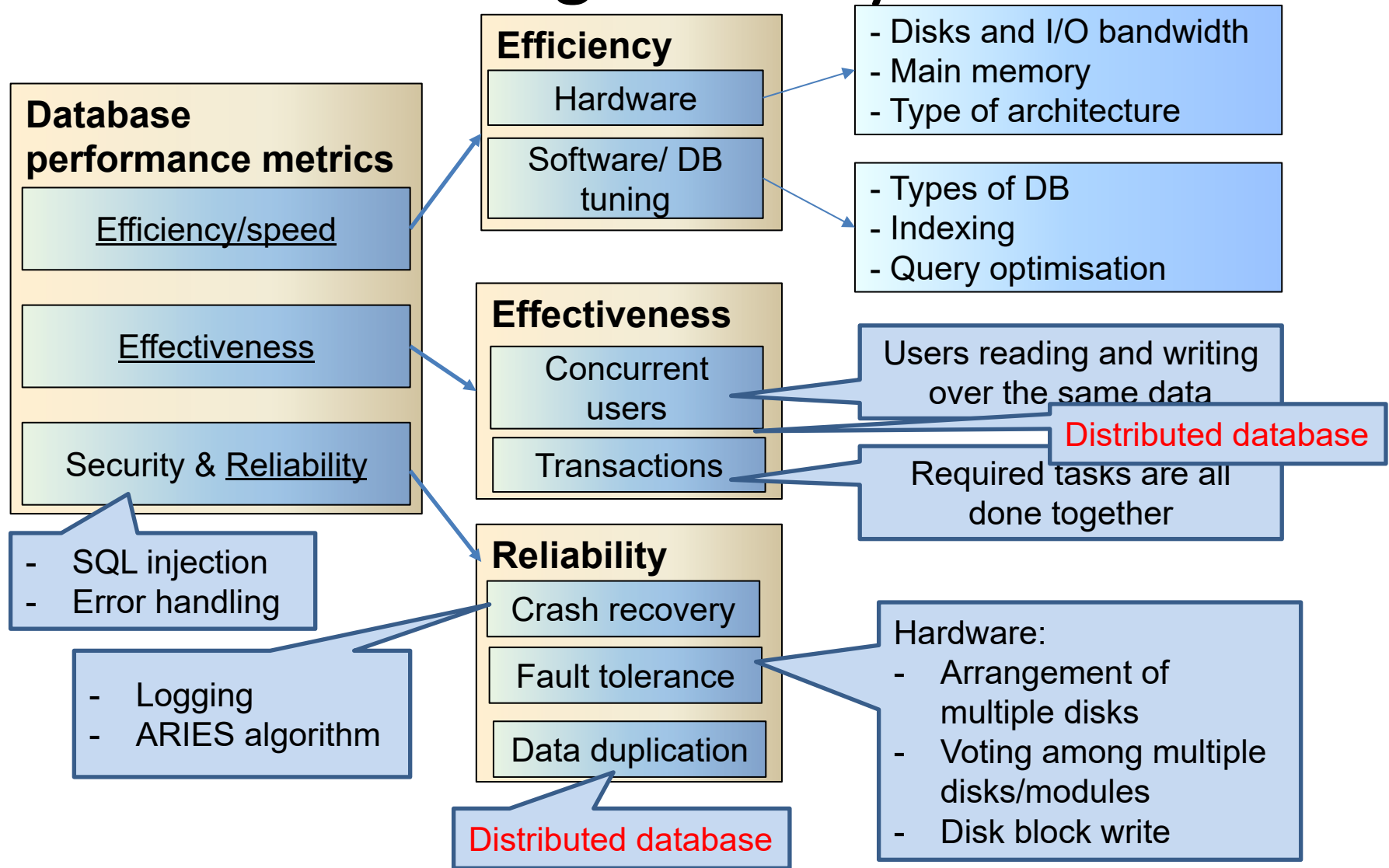
More real-life applications/examples where high availability is important, can sometimes tolerate some inconsistency?

Time for a poll –

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Core Concepts of Database management system



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<https://www.unimelb.edu.au/sls/students/ess>

Q/A